

HDOC Day-29

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Batch	15

Problem Statement

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge 2 arrays.
4. Exit.

Algorithm

1. Display the menu and read the user's choice.
2. Choice 1: read an array and count even, odd, positive, negative and zero values.
3. Choice 2: read an array and search key using linear search.
4. Choice 3: read two arrays and copy both into a merged array.
5. Choice 4: exit. Display an error for an invalid choice.
6. Repeat until Exit is selected.

C Program

```
#include <stdio.h>

int main() {
    int ch, n, i, key, pos;
    int a[100], b[100], c[200], n1, n2;
    int even, odd, positive, negative, zero;

    do {
        printf("\n1. Count elements\n");
        printf("2. Linear search\n");
        printf("3. Merge 2 arrays\n");
        printf("4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &ch);

        switch(ch) {
            case 1:
                printf("Enter size: ");
                scanf("%d", &n);
                printf("Enter elements: ");
                for(i=0; i<n; i++) scanf("%d", &a[i]);

                even=odd=positive=negative=zero=0;
                for(i=0; i<n; i++) {
                    if(a[i]%2==0) even++;
                    else odd++;

                    if(a[i]>0) positive++;
                    else if(a[i]<0) negative++;
                    else zero++;
                }
                printf("Even = %d\nOdd = %d\n", even, odd);
                printf("Positive = %d\nNegative = %d\nZero = %d\n",
                    positive, negative, zero);
                break;

            case 2:
                printf("Enter size: ");
                scanf("%d", &n);
                printf("Enter elements: ");
                for(i=0; i<n; i++) scanf("%d", &a[i]);

                printf("Enter element to search: ");
```

```

scanf("%d", &key);
pos=-1;
for(i=0; i<n; i++) {
    if(a[i]==key) {
        pos=i;
        break;
    }
}
if(pos!=-1)
    printf("Element found at position %d\n", pos+1);
else
    printf("Element not found\n");
break;

case 3:
    printf("Enter size of first array: ");
    scanf("%d", &n1);
    printf("Enter first array: ");
    for(i=0; i<n1; i++) scanf("%d", &a[i]);

    printf("Enter size of second array: ");
    scanf("%d", &n2);
    printf("Enter second array: ");
    for(i=0; i<n2; i++) scanf("%d", &b[i]);

    for(i=0; i<n1; i++) c[i]=a[i];
    for(i=0; i<n2; i++) c[n1+i]=b[i];

    printf("Merged array: ");
    for(i=0; i<n1+n2; i++) printf("%d ", c[i]);
    printf("\n");
    break;

case 4:
    printf("Program exited.\n");
    break;

default:
    printf("Invalid choice.\n");
}
} while(ch!=4);

return 0;
}

```

Test-Case Table

Choice	Input	Expected Result
1	-2 0 3 4 -5	Even=3, Odd=2, Positive=2, Negative=2, Zero=1
2	10 20 30 40; key=30	Found at position 3
3	1 2 3 and 4 5	Merged: 1 2 3 4 5
4	-	Program exits

Sample Output

Operation 1 - Count Elements

```
Enter choice: 1
Enter size: 5
Enter elements: -2 0 3 4 -5
Even = 3
Odd = 2
Positive = 2
Negative = 2
Zero = 1
```

Operation 2 - Linear Search

```
Enter choice: 2
Enter size: 4
Enter elements: 10 20 30 40
Enter element to search: 30
Element found at position 3
```

Operation 3 - Merge Arrays

```
Enter choice: 3
Enter size of first array: 3
Enter first array: 1 2 3
Enter size of second array: 2
Enter second array: 4 5
Merged array: 1 2 3 4 5
```

Operation 4 - Exit

```
Enter choice: 4
Program exited.
```